Downloaded from orbit.dtu.dk on: Nov 07, 2026



DTU Library

A Systematic Methodology for Design of Emulsion Based Chemical Products

Mattei, Michele; Kontogeorgis, Georgios; Gani, Rafiqul

Publication date:
2012

Document Version
Publisher's PDF, also known as Version of record

Link back to DTU Orbit

Citation (APA):
Mattei, M., Kontogeorgis, G., & Gani, R. (2012). *A Systematic Methodology for Design of Emulsion Based Chemical Products*. Poster session presented at 11th International Symposium on Process Systems Engineering, Singapore.

General rights
Copyright and moral rights for the publications made accessible in the public portal are retained by the authors and/or other copyright owners and it is a condition of accessing publications that users recognise and abide by the legal requirements associated with these rights.

- Users may download and print one copy of any publication from the public portal for the purpose of private study or research.
- You may not further distribute the material or use it for any profit-making activity or commercial gain
- You may freely distribute the URL identifying the publication in the public portal

If you believe that this document breaches copyright please contact us providing details, and we will remove access to the work immediately and investigate your claim.

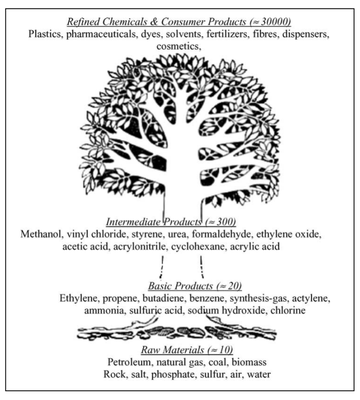
A Systematic Methodology for Design of Emulsion Based Chemical Products

Michele MATTEI, Georgios M. KONTOGEORGIS, Rafiqul GANI

Department of Chemical and Biochemical Engineering, Technical University of Denmark
Søltofts Plads, Building 229, DK-2800, Kgs. Lyngby, Denmark,

1. CHEMICAL PRODUCT DESIGN AND EMULSIONS

- Chemical product design aims to find a product exhibiting a set of deisrable or specified behavior
- Chemical industry is shifting from commodities towards higher value added products
- Higher value added products gain their value from a molecular or micro-structure
- Emulsions are largely used as commercial products: food, cosmetic, household and health-care industries
- Surfactant and emulsion properties may require dedicated models



2. A SYSTEMATIC METHODOLOGY

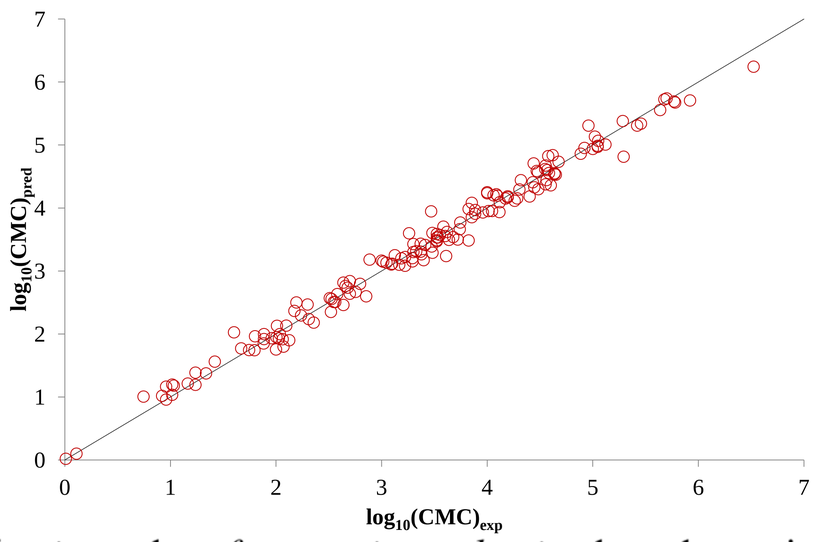
Step	Input	Performed Action	Output
Step 1	Information about the product	Understanding of needs, translation into target properties in terms of constraints to match	List of main needs, secondary needs and target properties with boundaries
Step 2	List of main needs	Find chemicals suitable as active ingredient(s), select the most advantageous and search/calculate main properties	Candidate active ingredient with its main properties
Step 3	Candidate active ingredient with its main properties	Find chemicals suitable as dispersed phase solvents, select the most advantageous and search/calculate main properties	Candidate dispersed phase solvent with its main properties
Step 4	Candidate active ingredient and dispersed phase solvent with their main properties	Find chemicals suitable as continuous phase solvents, select the most advantageous and search/calculate main properties	Candidate continuous phase solvent with its main properties
Step 5	Candidate active ingredient dispersed and continuous phase solvents with their main properties	Find chemicals suitable as emulsifiers, select the most advantageous and search/calculate main properties	Candidate emulsifier with its main properties
Step 6	List of secondary needs, candidate dispersed and continuous phase solvents with their main properties	Find chemicals suitable as additives, select the most advantageous and search/calculate main properties	Candidate additives with their main properties
Step 7	List of constraints on target properties, all ingredients	Find overall composition of selected ingredients	Product to be experimentally verified (Stage-2)

3. SELECTION CRITERIA

- *Constraints*
  - Low toxicity → Lethal concentration: $LC_{50} > 3{,}16$
  - Safety → Flash point: FP > 60°C
  - Legislation → Some chemicals are not allowed
  - Critical temperatures → Melting point, boiling point, peculiar critical temperatures of surfactants
  - Stability → Surfactant above critical micelle concentration
- *Selection criteria*
  - Optimize the effectiveness (it differs ingredient by ingredient and it is estimated mainly with knowledge base)
  - Minimize the cost

4. SURFACTANT PROPERTY MODELING

- *Critical micelle concentration* (158 data)
  - Non-ionic surfactants, T = 25°C
  - M&G GC⁺ method: $f(X) = \sum_i N_i C_i + w \sum_j M_j D_j + z \sum_k O_k E_k$
  - $R^2 = 0{,}99$



- *Cloud point* and *surface tension reduction* have been similarly modeled with high fitting quality
- Modeling of *ionic surfactant* properties will be the next step

5. A CONCEPTUAL CASE STUDY

- UV sunscreen, emulsified form
- Proposed formulation to be experimentally validated



Avobenzone	0,99%		
Octyl Salycilate	1,22%	Octyl Esaethylene Oxide	0,09%
Zinc Oxide	8,56%	α-Tocopherol	0,34%
Butyl Acetate	32,4%	Heptylparaben	0,59%
Water	54,9%	*laevo*-Menthol	0,91%

6. CONCLUSIONS

- A systematic methodology for design of emulsion-based products has been developed
- Its application has been illustrated through the conceptual case study of an emulsified UV sunscreen
- The development of dedicated models for pure compound (surfactant) as well as mixture (emulsion) properties is fundamental
- A second stage, model-experiment based further refinement and/or validation has been considered and needs to be developed
- The work-flow will be implemented into a software to allow virtual product design

MAIN REFERENCE

- Cussler E.L., Moggridge G.D., Chemical Product Design, 2011
- Conte E. et al., AIChE Journal; 2011, 58(1): 173-189
- Gani R., Comput. Chem. Eng.; 2004, 28 (12): 2441-2457
- Couteau C. et al, J. Derm. Sc.; 2009, 30 (10): 159-161